

BRD Coverage Map — AI Virtual Building Tour

Every BRD area mapped to the running code in `virtual-building-tour/`. Verified by an automated end-to-end test: **21/21 pipeline steps pass**, producing a real `.glb` model and `.mp4`.

- Working** — functional now
- Basic/real** — works, built to be upgraded
- Stub** — placeholder, heavy step deferred

Accounts, roles & security — BRD 7.1, 16

BRD	REQUIREMENT	STATUS	WHERE
AUTH-001..004	Register, email-verify gate, login, sessions	WORKING	routers/auth.py
4.1 / BR-008	Roles (super/company admin, designer, viewer) + RBAC	WORKING	security.py, require_roles
16	Password hashing, signed tokens, audit trail	WORKING	security.py, utils.audit
AUTH-006	Google/Microsoft SSO readiness	READY	identity model isolated

Projects & dynamic forms — BRD 7.2–7.5

BRD	REQUIREMENT	STATUS	WHERE
PRJ-001..005	Create, list, view, delete, status lifecycle	WORKING	routers/core.py
BLD-001..004	Apartment/villa selection + dynamic building form	WORKING	core.upsert_building
APT/VIL	Floors, units, rooms, villa site features	WORKING	models.py, core floors/rooms
6.1	Project status lifecycle (11 states)	WORKING	models.PROJECT_STATUSES

Upload & AI analysis — BRD 7.6, 8

BRD	REQUIREMENT	STATUS	WHERE
UPL-001..003	Upload JPG/PNG/PDF, validate type/size, checksum	WORKING	core.upload_plan
8.1	Detect walls, rooms, doors/windows from the plan	BASIC	services/analysis.py (OpenCV)
8.2	Per-element confidence + recorded model version	WORKING	analysis.py, AnalysisJob
8.3	Quality warnings (blocking / non-blocking)	WORKING	analysis.quality_warnings

2D verify & 3D generation — BRD 7.7, 9

BRD	REQUIREMENT	STATUS	WHERE
EDT-001,004	Overlay preview + scale calibration	PREVIEW	frontend renderVerify
EDT-002	Drag add/move/delete of elements	PHASE 3	canvas editor to add
EDT-007 / BR-006	Approve = locked, versioned source of truth	WORKING	core.approve_plan
9.1–9.2	Empty building: walls, floor, openings, default theme	REAL	generation3d.py + glb.py
9	Multi-floor apartment stacking → .glb	REAL	generation3d.generate

Tour, video & viewer — BRD 10, 16

BRD	REQUIREMENT	STATUS	WHERE
TOUR-001,002	Auto walkthrough route + editable scenes	WORKING	services/tour.py
16 (interactive)	Browser 3D tour: orbit/walk, floors, play, share link	WORKING	frontend/viewer.html (Three.js)
TOUR-007 / BR-013	Shareable link with access level	WORKING	core.shared_tour
TOUR-005,006	MP4 render, 16:9 / 9:16, 1080p	STUB	services/render.py (FFmpeg)

Ops, data & platform — BRD 12, 13, 15

BRD	REQUIREMENT	STATUS	WHERE
12	Admin: jobs, users, storage, audit log	WORKING	routers/admin.py
7.10	Notifications (done / failed / review-required)	WORKING	core notifications
13.1–13.2	All core entities + versioning rules	WORKING	models.py (15 tables)
15 / NFR	Scalable workers (Celery), Postgres, object store	READY	docker-compose, config swap

What "real vs stub" means here. The whole pipeline runs end-to-end today. The three items to upgrade are the deliberately-hard parts from the plan: **AI accuracy** (swap OpenCV heuristics for trained YOLO + segmentation + OCR), the **full 2D drag-editor** (Phase 3), and **video rendering** (replace the FFmpeg placeholder with a Blender camera-path render on a background worker, Phase 6). Everything they plug into — endpoints, data model, versioning, viewer — already exists.

Run it: `cd virtual-building-tour && ./run.sh` → open `http://localhost:8000` → log in `demo@vbt.local / demo1234` → create project → upload → analyse → approve → generate 3D → open viewer.

Generated from the running codebase. Disclaimer (BRD): outputs are visual representations only — not certified architectural, structural or legal documents.